# Studio North: customer economics lab

## Are we moving in the right direction as a company?

**Compare acquisition efficiency, contribution value and cash recovery across cohorts.**

A monthly creative-work membership for independent professionals: templates save time, guided learning builds skills, and workshops add value.

24 joining cohorts: January 2024–December 2025.
Monthly renewals; cutoff January 1, 2026 (exclusive).

| | |
|---|---|
| CRM | Renewals |
| Transactions | Product events |
| Feedback | Campaign spend |
| Source credit | Assignments |

# How to read the slide colors

| | |
|---|---|
| **Concept** | Definitions and distinctions to use when interpreting the data. |
| **Case application** | Studio North and its connected customer records. |
| **Evidence interpretation** | Observed results, predictions, and what each can establish. |
| **Group decision** | One focused judgment, discussed with your group. |

# Three ideas guide the entire lab

## 1. Connected customer evidence

The Universal Customer View classifies each dataset's fields as Identifiers, Engagement, Relationship or Perception. It reveals what is recorded and what is missing.

## 2. Customer economics over time

Joining cohorts distinguish customer age from calendar time. Retention, contribution and acquisition cost determine whether relationships create value.

## 3. Incremental marketing value

A randomized campaign comparison asks what changed because of the campaign. A high customer LTV alone cannot answer that question.

**AI proposes the work. Saved SQL, data checks, held-out outcomes and your judgment make the work reviewable.**

# Every dataset gets its own Universal Customer View

Example: people.csv contains one record per person, including prospects who never buy.

Table 1. Field classification for people.csv

| Identifiers | Engagement | Relationship | Perception |
|---|---|---|---|
| person_id<br>age_band<br>household_type<br>region | — | record_created_at<br>attributes_recorded_at | primary_need |

The stated need is a reported intention. Profile timestamps document when the relationship information became available.

**Repeat this field-level table for all eight exports, then open the Archify map of keys, grains and connections.**

# Group decision 1: who needs which onboarding route?

Studio North can offer a ready-to-use template or a guided lesson. Which recorded grouping should inform these onboarding routes?

Table 2. Two candidate grouping rules

| Grouping | Two recorded groups | What the field describes |
|---|---|---|
| Age band | 18–34 / 35+ | A demographic characteristic. |
| Stated need | Save time / Build skill | The goal stated before acquisition. |

**Choose one grouping and explain why it is relevant to this onboarding objective.**

**Use the conceptual journey maps to propose the routes. Later, use the SQL results to test whether the grouping explains value.**

# Turn the connected view into a reproducible query

### Inspect raw views

Keep source columns and row meanings.

### Define the customer clock

First paid month sets cohort age index 0.

### Build SQL outputs

Save the exact SQL and ordered results.

### Reconcile the totals

No duplicate money; no missing controls.

Figure 1. From source views to reconciled cohort evidence

## Prompt to try

“Show the SQL that calculates retained customers and contribution by joining month and customer age. Explain the denominator and preserve unobserved future ages as missing.”

**A SQL script makes a calculation inspectable; the data definitions determine whether it answers the right question.**

# A cohort follows the same original customers

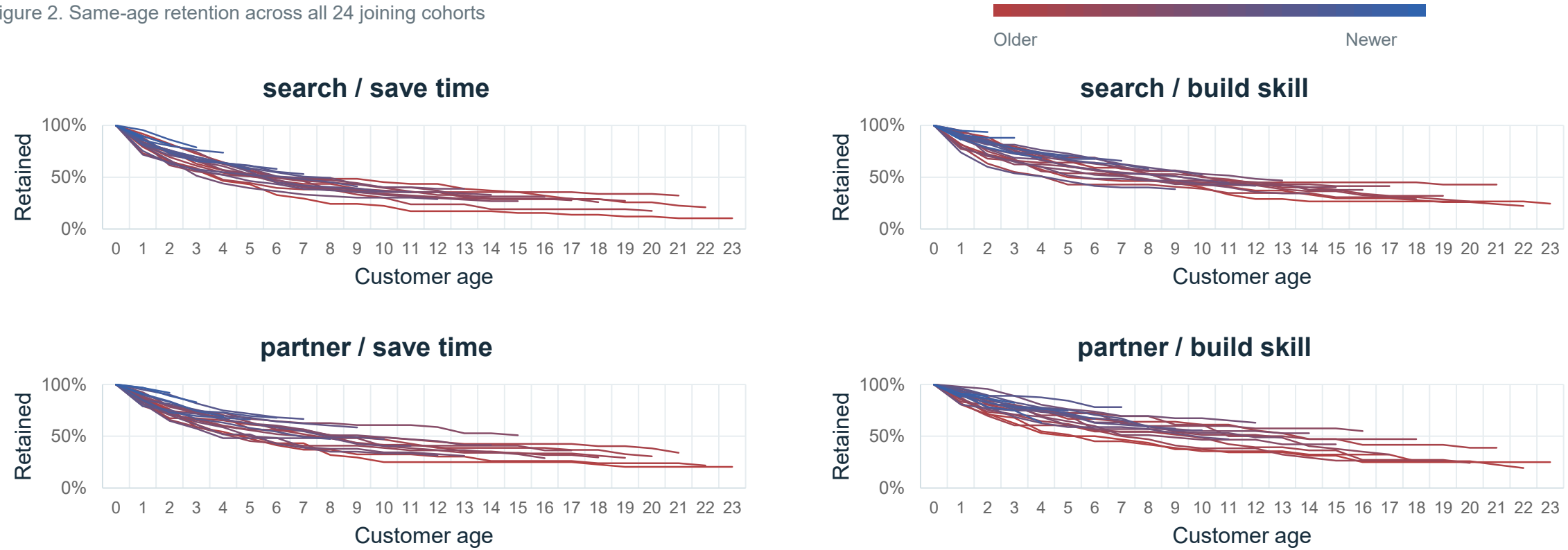
Table 3. Retained share by joining cohort and customer age (months)

| Cohort | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024-01 | 100% | 86% | 75% | 64% | 55% | 50% | 45% | 41% | 35% | 33% | 30% | 26% | 25% | 25% | 24% | 23% | 22% | 22% | 21% | 20% | 20% | 20% | 20% | 19% |
| 2024-02 | 100% | 86% | 76% | 67% | 59% | 52% | 47% | 45% | 44% | 42% | 40% | 39% | 37% | 36% | 33% | 31% | 30% | 29% | 27% | 25% | 25% | 23% | 21% | — |
| 2024-03 | 100% | 89% | 78% | 72% | 67% | 59% | 56% | 54% | 52% | 49% | 47% | 46% | 44% | 43% | 42% | 42% | 41% | 41% | 40% | 39% | 38% | 37% | — | — |
| 2024-04 | 100% | 78% | 66% | 60% | 55% | 51% | 48% | 46% | 45% | 42% | 40% | 35% | 33% | 33% | 31% | 30% | 28% | 28% | 28% | 26% | 24% | — | — | — |
| 2024-05 | 100% | 82% | 70% | 65% | 60% | 58% | 55% | 50% | 47% | 45% | 41% | 41% | 39% | 35% | 34% | 32% | 31% | 30% | 30% | 29% | — | — | — | — |
| 2024-06 | 100% | 85% | 74% | 67% | 59% | 55% | 54% | 51% | 49% | 46% | 44% | 41% | 39% | 38% | 36% | 36% | 34% | 34% | 32% | — | — | — | — | — |
| 2024-07 | 100% | 88% | 76% | 70% | 65% | 62% | 58% | 54% | 48% | 43% | 42% | 41% | 40% | 39% | 38% | 37% | 35% | 34% | — | — | — | — | — | — |
| 2024-08 | 100% | 86% | 76% | 69% | 64% | 60% | 56% | 53% | 51% | 50% | 47% | 45% | 43% | 40% | 39% | 38% | 37% | — | — | — | — | — | — | — |
| 2024-09 | 100% | 88% | 79% | 72% | 65% | 60% | 56% | 51% | 51% | 47% | 46% | 44% | 43% | 40% | 40% | 39% | — | — | — | — | — | — | — | — |
| 2024-10 | 100% | 87% | 76% | 70% | 63% | 57% | 53% | 51% | 48% | 46% | 45% | 43% | 42% | 40% | 39% | — | — | — | — | — | — | — | — | — |
| 2024-11 | 100% | 88% | 78% | 70% | 66% | 61% | 56% | 53% | 50% | 48% | 45% | 44% | 42% | 41% | — | — | — | — | — | — | — | — | — | — |
| 2024-12 | 100% | 88% | 75% | 65% | 58% | 53% | 49% | 46% | 44% | 43% | 42% | 42% | 40% | — | — | — | — | — | — | — | — | — | — | — |
| 2025-01 | 100% | 82% | 71% | 63% | 58% | 53% | 52% | 49% | 48% | 45% | 43% | 42% | — | — | — | — | — | — | — | — | — | — | — | — |
| 2025-02 | 100% | 83% | 76% | 73% | 70% | 64% | 59% | 55% | 52% | 49% | 47% | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 2025-03 | 100% | 81% | 71% | 64% | 61% | 58% | 54% | 52% | 50% | 48% | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 2025-04 | 100% | 86% | 81% | 73% | 69% | 64% | 61% | 57% | 53% | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
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| 2025-06 | 100% | 88% | 82% | 75% | 71% | 67% | 65% | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 2025-07 | 100% | 87% | 78% | 73% | 70% | 67% | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 2025-08 | 100% | 88% | 80% | 75% | 69% | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 2025-09 | 100% | 94% | 88% | 83% | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 2025-10 | 100% | 93% | 89% | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 2025-11 | 100% | 88% | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 2025-12 | 100% | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |

Read across for one cohort over time, or down for different cohorts at the same age. Future ages stay missing.

# Group decision 2: is improvement visible within groups?

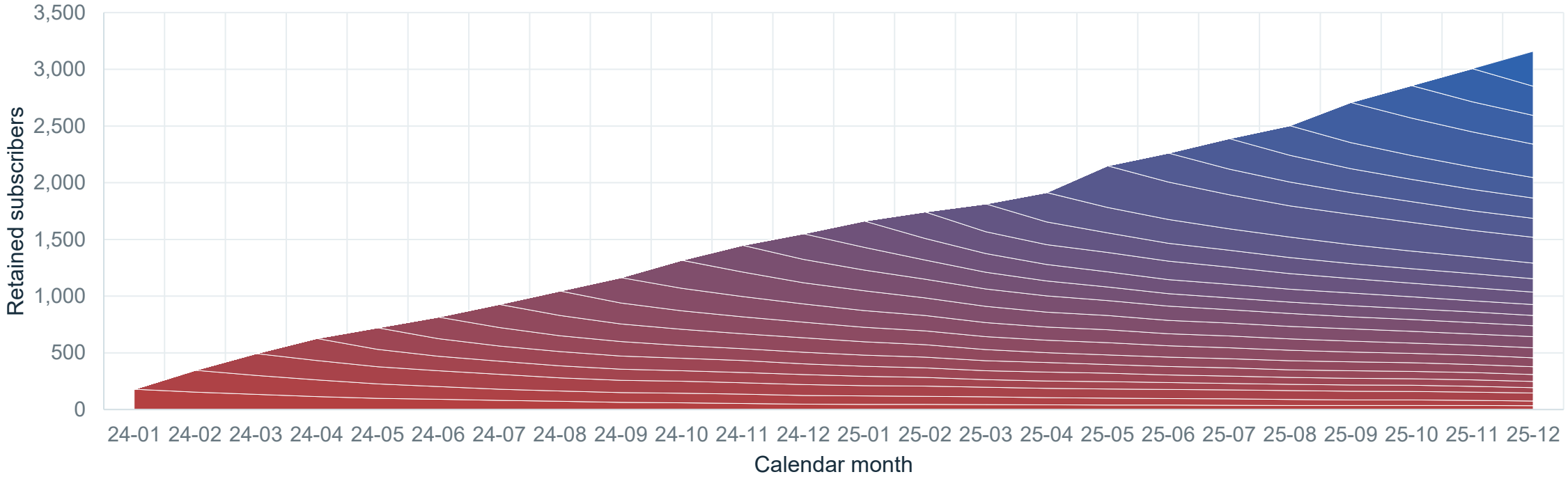
Figure 2. Same-age retention across all 24 joining cohorts




**Do newer cohorts retain better at the same age? Is the progress broad or concentrated?**

# Joining cohorts accumulate into the company

Figure 3. Retained subscribers by joining cohort and calendar month




**Each band is one joining cohort. The height is the retained customer base, not cumulative acquisitions.**

# Group decision 3: is one month enough to decide?

Each campaign randomized 250 prospects to treatment and 250 to its own control. Follow-up lasts about one month.

Table 4. Experimental contribution after acquisition spending

| Test | Follow-up | Incremental contribution after media / prospect | Approx. 95% interval |
|---|---|---|---|
| search / save time | 31 days | -$0.65 | -$2.29 to $0.98 |
| partner / build skill | 30 days | -$1.31 | -$3.13 to $0.51 |

**Would you scale, stop, or keep measuring? What evidence would change your decision?**

**Cohort payback takes months. Predicted customer LTV cannot replace longer experimental follow-up.**

# Model renewal and revenue before forecasting LTV

## Retention model

A hierarchical shifted Beta-Geometric model represents persistent differences in churn risk. Cohort and group information share strength where histories are short.

## Revenue model

A Gamma model describes positive revenue among retained customers. Combine this forecast with retention, variable costs and observed contribution.

## Validation

Compare chronological holdout predictions with a simple baseline and an Orduz-style neural network. Keep uncertainty visible; a fitted curve is not a guarantee.

**Use a common 12-month horizon. Net LTV subtracts CAC; cash payback tracks undiscounted recovery.**

**Run the locked NumPyro/JAX models on CPU with uv. Describe the model before showing its results.**

# Hand your team a decision and a reproducible report

## Are successive cohorts becoming more valuable relative to acquisition cost?

Table 5. The executive handoff

| Report component | What your team should receive |
|---|---|
| Executive summary | The business, a supported verdict, a proposed action and what would change it. |
| Observed and forecast evidence | Same-age retention, revenue, contribution, CAC and cash payback. |
| Model appendix | Model story, parameters, priors, uncertainty and chronological validation. |
| Data and reproducibility appendix | UCV tables, Archify key map and saved SQL; saved manifests accompany the report. |

**Bonus challenge: forecast 2026 outcomes for these 24 cohorts with another CPU-capable model. Submit for instructor-held evaluation.**

**The report skill carries the style and appendices. Label every figure and table; preserve the evidence trail.**